KARAN KAHER

Data Engineer — Data Platform Engineer — AI & Data

Delhi, India | +91 8178972701 | kaherkaran@gmail.com | linkedin.com/in/karan-kaher-55645a148

Professional Summary

Data Engineer with 5+ years of experience building and optimizing data pipelines, cloud data warehouses, and BI reporting systems for enterprise clients across Insurance, Finance, and Retail/FMCG. Known for diagnosing and fixing performance bottlenecks in large-scale SQL systems and for turning ambiguous business requirements into reliable, production-grade data solutions. Actively deepening expertise in applied Generative AI system design; Oracle GenAI-certified with a perfect 100% score.

Technical Skills

Languages	Python, SQL, PySpark
Data Engineering	Apache Airflow, Apache Kafka, DBT, SSIS, Databricks
Data Modeling / Warehousing	Snowflake, AWS Redshift, SQL Server, DB2, Dimensional Modeling, Star Schema, ETL/ELT Design
Cloud Platforms	Microsoft Azure, AWS (Redshift, S3, EC2)
DevOps & Infrastructure	Docker, Kubernetes, Terraform, Ansible, Jenkins, GitLab CI/CD, GitHub Actions, Argo CD
Generative AI	RAG, LLM Agents, Prompt Engineering, Vector Search, LLM Concepts, Claude Code
BI & Visualization	Power BI, Tableau, SSRS, SSAS

Professional Experience

Software Engineer – Data & Analytics — **Feb 2021 – Present**

Damco Solutions Private Limited (IT Services & Consulting) — *Remote, India*

- Diagnosed slow-running SSRS reports by analyzing SQL Server execution plans, then rewrote 40+ queries – replacing correlated subqueries with CTEs/window functions, adding covering and composite indexes, and removing implicit data-type conversions that were blocking index seeks – cutting average execution time by up to 60% across 92 reports over 3 years.
- Restructured the underlying claims/premium schema for AML compliance reporting after tracing a recurring data-mismatch bug to inconsistent join grain between fact and dimension tables, eliminating duplicate-flagging errors in suspicious-transaction reports.
- Redesigned SSIS ETL packages extracting policy, claims, premium, and finance data from multiple source systems, replacing full-table reloads with incremental/delta extraction logic, reducing nightly batch runtime by roughly 30%.
- Authored a company-wide SSRS Reporting SOP codifying query optimization patterns, dataset caching strategy, subscription scheduling, and parameter design; adopted as the team's standard, cutting new-developer ramp-up time by roughly 25%.
- Built PySpark transformation jobs on Databricks to replace a legacy single-threaded SQL batch process for Finance-domain reconciliation, and delivered the resulting Power BI dashboards tracking KPIs and loss ratios for actuarial stakeholders.
- Partnered directly with actuarial and finance stakeholders across multiple geographies to convert ambiguous reporting requests into parameterized, self-service Power BI dashboards, reducing recurring ad-hoc reporting requests.

Key Projects

Retail/FMCG – Cloud Data Integration & Analytics Platform

- **Situation/Task:** A retail/FMCG client's sales and supply-chain data was fragmented across multiple regional source systems, forcing analysts to manually reconcile spreadsheets before any reporting could happen.
- **Action:** Designed Airflow DAGs to orchestrate multi-source extraction on a scheduled cadence, built dbt models (with source/schema tests) to standardize and transform the data, and modeled a star schema (fact_sales, dim_product, dim_region, dim_time) in Snowflake for downstream analytics.
- **Result:** Replaced manual spreadsheet reconciliation with a single automated pipeline, improving data freshness from a weekly manual refresh to daily automated loads and giving analysts a trusted, query-ready dataset.

Cloud Data Warehouse Migration & Optimization (AWS Redshift / Snowflake)

- **Situation/Task:** Legacy on-premise reporting workloads were becoming slow and expensive to maintain as data volumes grew, and stakeholders needed faster turnaround on ad-hoc queries.

- **Action:** Redesigned ETL jobs using Airflow and dbt to land data into a cloud warehouse, restructured wide fact tables into properly normalized dimensional models, and tuned physical design (distribution/sort keys in Redshift, clustering keys in Snowflake) to match common query patterns.
- **Result:** Cut average query runtime by roughly 35% and reduced the on-premise infrastructure footprint the client had to maintain.

Insurance – Real-Time Claims Processing & Fraud Monitoring

- **Situation/Task:** Fraudulent claims were only being caught through nightly batch jobs, giving fraud analysts a day-plus lag between a suspicious claim being filed and being flagged.
- **Action:** Built a Kafka-based event pipeline to stream claim events as they were filed, and wrote Spark Structured Streaming jobs applying rule-based anomaly checks (claim-amount thresholds, claim-frequency-per-policyholder, geographic/time-of-day outliers) to score each event in near real time; built a companion event-driven pipeline moving auto-claims from intake through adjudication-ready state with automated data-quality checks at each stage.
- **Result:** Cut the fraud-detection lag from a full batch cycle to near real time, surfacing flagged claims to the fraud team through analytical dashboards before payout.

Generative AI – Claims Assistant & Knowledge Intelligence Agent

- **Situation/Task:** Business and sales teams were spending significant manual effort searching through policy/claims documents and competitor market research to answer routine questions.
- **Action – persona & workflow:** Designed the assistant as a "Senior Claims Due Diligence Analyst" persona with a defined end-to-end workflow – classify the incoming query, retrieve supporting evidence, cross-check evidence against policy rules, then produce a recommendation – rather than a single free-form prompt.
- **Action – RAG pipeline:** Chunked source documents with semantic, overlap-aware splitting, generated embeddings, and implemented vector similarity search with a reranking step to retrieve the most relevant passages per query.
- **Action – output & guardrails:** Constrained every response to a fixed JSON schema (`finding`, `confidence`, `supporting_sources`, `recommended_action`) so downstream tools could consume it directly, and enforced behavioral rules – cite source documents for every claim, decline and flag for human review when retrieval confidence falls below threshold, and escalate out-of-scope queries rather than guessing.
- **Result:** Delivered a working internal agent that returns structured, source-cited, auditable recommendations instead of free-text guesses, reducing manual document lookup time for routine claims/market-research queries.

Insurance & Finance – Enterprise BI / KPI Reporting Platform (Power BI)

- **Situation/Task:** Leadership had no single consolidated view of claims, underwriting, and financial KPIs, relying instead on multiple disconnected spreadsheets maintained by different teams.
- **Action:** Designed a shared semantic data model with row-level security by business unit, built parameterized dashboards with drill-through from summary KPIs into transaction-level detail, and configured scheduled automated refresh.
- **Result:** Consolidated fragmented reporting into a single governed source of truth, cutting the manual effort previously spent assembling recurring KPI reports.

Education

B.Tech, Computer Science & Engineering **2016 – 2020**

Manav Rachna University *CGPA: 8.88*

Certifications

- **Oracle Generative AI Professional** – Oracle (100% Score)
- **Microsoft Azure AI Fundamentals (AI-900)** – Microsoft (Passed)
- **IBM Data Science Specialization** – IBM / Coursera (Completed)
- **MSBI – SQL Server BI Stack** – Udemy (Completed)

Extra-Curricular

- IEEE Active Member – Hackathons & Tech Competitions
- Zenith Semifinalist – Ace the Excellence, CDC